Pole equations from simultaneous congruent-number twists

14 November 2026

Abstract

Three equations forbid a negative odd valuation of their rational input at every odd prime. For prepared positive inputs less than one, the simultaneous-twist theorem of Jaswal supplies rational witnesses. The application checks the precise square classes, local congruences and scaling of both elliptic points, and treats the square input separately. Seven auxiliary rational coordinates give a quadratic presentation, with smaller coefficients than the previous pole system. The pole replacement does not reduce the coordinate count on its own.

1 The equations and their local implication

For $b \in \mathbb{Q}$, use rational e, p, u, v, q and require

$$e^2 + bp^2 = 1, \qquad uv = e(1-b), \qquad u^2 - v^2 = bq^2. \tag{1}$$

No nonzero restriction is imposed for the local implication.

Theorem 1.1. *If (1) has a rational solution and $b \neq 0$, then $v_\ell(b)$ is not a negative odd integer for any odd prime ℓ.*

Proof. Suppose $\nu = v_\ell(b) < 0$ is odd. The first equation has terms of different valuation parity whenever both are nonzero. It forces $v_\ell(e) = 0$ and $v_\ell(bp^2) > 0$ (with the latter interpreted as infinity when $p = 0$): a negative smaller valuation could not cancel, and a zero valuation on bp^2 is impossible by parity. Thus $e \neq 0$ and $v_\ell(uv) = v_\ell(e(1-b)) = \nu$, so both u, v are nonzero and their valuations have odd sum. In particular they are distinct. The valuation of $u^2 - v^2$ is then the smaller of two distinct even integers, and is even. The right side bq^2 is either zero or has odd valuation. Both possibilities contradict the nonzero even-valuation left side. □

2 The exact simultaneous-twist input

Write $E_r : y^2 = x^3 - r^2x$. We use the following statement, exactly the scope of Jaswal's Theorem 1.2 in Hexagon version 2 [1]: for coprime positive squarefree integers $\beta > 1$, $\theta \geq 1$, with $\beta \equiv \theta \equiv 1 \pmod 8$, each a square modulo every prime dividing the other, there is a positive squarefree integer r such that $E_r, E_{r\theta}$ both have an infinite-order rational point whose abscissa lies in $\beta\mathbb{Q}^{\times 2}$. The source gives analytic and algebraic rank one as well, but those stronger conclusions are not needed for the following application.

We retain this stated theorem as the input to the realization result. The application below is proved in full; it is not a claim that every part of the source's external arithmetic proof has been independently re-audited. No use is made of its separate contact-height or prime-pattern statements. In particular, the theorem does not assert that arbitrary different prescribed square classes share witnesses.

Theorem 2.1. *Assume the simultaneous-twist statement above. Suppose $b \in \mathbb{Q}$ has square reduced denominator and satisfies*

$$0 < b < 1, \qquad b \in \mathbb{Q}_2^{\times 2}, \qquad v_2(b) \geq 3.$$

Then (1) has a rational solution with $upq \neq 0$.

Proof. Write $b = \beta z^2$ and $1 - b = \theta c^2$, where β, θ are positive squarefree integers and $z, c \in \mathbb{Q}^\times$. More explicitly, if $b = A/B^2$ in lowest terms, then β is the squarefree part of A, and θ that of $B^2 - A$. They are coprime: a prime dividing both would divide A and B, which is impossible. At each odd prime dividing β, the value $1 - b$ has square unit residue, so θ is a square modulo that prime. Conversely at a prime dividing θ, b has square unit residue and β is a square. These primes cannot divide the reduced denominator.

Because b is a 2-adic square, $v_2(b)$ is even and is at least four. The square denominator is therefore odd in reduced form. The squarefree part β is odd and is $1 \pmod 8$, as follows from the square unit part of b. Also $1-b \equiv 1 \pmod{16}$ in $\mathbb{Z}_2$, so its odd squarefree part θ is $1 \pmod 8$.

If $\beta = 1$, put $e = 0$, $p = 1/z$, $u = z$, $v = 0$ and $q = 1$. This proves the claim directly and does not apply a theorem requiring $\beta > 1$. Otherwise apply the simultaneous-twist input to β, θ. Scale the point on $E_{r\theta}$ by $(x, y) \mapsto (c^2x, c^3y)$, obtaining a point on $E_{r(1-b)}$. Both abscissas remain in $\beta\mathbb{Q}^{\times 2} = b\mathbb{Q}^{\times 2}$. Write the resulting points as (ba_1^2, y_1) on E_r and (ba_2^2, y_2) on $E_{r(1-b)}$, with $a_1a_2y_1y_2 \neq 0$. Their ordinates are nonzero because the points have infinite order. Set

$$e = \frac{r}{ba_1^2}, \quad p = \frac{y_1}{b^2a_1^3}, \quad u = \frac{a_2}{a_1}, \quad v = \frac{e(1-b)}{u}, \quad q = \frac{y_2}{b^2a_1^3u^2}.$$

The first curve equation, divided by $b^3a_1^6$, gives $bp^2 = 1 - e^2$. The second, divided by $b^3a_1^6u^2$, gives $bq^2 = u^2 - e^2(1-b)^2/u^2 = u^2 - v^2$. The middle equation is the chosen definition of v. All divisions are by quantities just shown nonzero, and $upq \neq 0$. □

3 Quadratic presentation and coefficient comparison

Retain two additional variables p_2, q_2 and impose five quadrics:

$$p_2 = p^2, \quad q_2 = q^2, \quad e^2 + bp_2 = 1, \quad uv = e(1-b), \quad u^2 - v^2 = bq_2. \tag{2}$$

There are seven auxiliary coordinates, the same number as before. The replacement changes three rows per detector while keeping the two square definitions. It uses no fixed factor 17, and every displayed coefficient has absolute value one. The new prepared range is $(0, 1)$; in the abc quartic the actual pole input tends to zero on the selected real neighborhoods, so this range can be imposed before prime selection. The old extra squareclass conditions at 3 and 17 may be retained but are not consumed by Theorem 2.1.

At a prime with positive odd contact order, the unchanged trace equations force a nonzero pole input of odd valuation. Theorem 1.1 therefore gives the same strict local inequalities as the older pole system. No assertion about a zero pole input is needed away from these consumed primes. The new global zero set is not claimed identical to the old pole equations for every b; soundness and completeness of the whole construction are proved separately.

In the unchanged 81-coordinate reference presentation the full quartic has exact height $\max(4153456, B^2 + 630511)$. In the compressed 79-coordinate presentation the reference equations, not the pole coefficients, dominate the height. Both statements are verified after expanding and collecting all coefficients in their respective complete tables. They do not give an effective sufficient defining shift.

References

[1] M. Jaswal, *A proof that $\mathbb{Z}$ is Diophantine in $\mathbb{Q}$ through congruent-number twists*, Hexagon `2611.00331v2`, 12 November 2026, Theorem 1.2, printed p. 4. https://hexagonmath.org/pdf/2611.00331v2.

[2] *A 79-variable quartic under rational abc*, project companion, `quartic_abc79_20261114_v27.tex`.